Pre-Requisite:

Requirement Gathering and Wireframe Design

Reference Videos:

- https://www.youtube.com/watch?v=6t_dYhXyYjI
- https://www.youtube.com/watch?v=CWpiKEbRsz4
- https://www.youtube.com/watch?v=PeGfX7W1mJk&pp=ygUJI3R1dG9hcHBz

Setting Up Flutter Development Environment

Reference Videos:

- https://www.youtube.com/watch?v=NacSPZyFjZY
- https://www.youtube.com/watch?v=_DBd-hqYR8Y&list=PLlvhNpz1tBvH4Wn8rMjtscK3l2pXnC9aN&index=2&pp=iAQB
- https://www.youtube.com/watch?v=DDR2kHwbdDs&list=PLUhfM8afLE_OaJOVXCboP9PRuhvirFqgk&index=4&pp=iAQB

Dart official Docs: https://dart.dev/
Flutter official Docs: https://docs.flutter.dev/

Entire Dart Tutorial -
https://www.youtube.com/watch?v=2qqplOILWoc&list=PLhyraTKIsw58r8DGaceGeTZKVcSXPabd0

Flutter full course - https://www.freecodecamp.org/news/learn-flutter-full-course/
The complete Dart and Flutter course - https://www.youtube.com/watch?v=CzRQ9mnmh44

Week 1: Introduction to Flutter
Lab-1:
Topics:
Flutter Architecture
Dart Programming Language

Learning Resources - Introductory Dart Video (First 2 hrs.)
https://youtu.be/ka2XxumyR0w

Practical Task –

You will build a Dart program that:

1. Collects user input (name, age, and favourite colour).
2. Stores this input in variables.
3. It uses conditional statements to print personalised messages based on age.
4. Organises the collected data into collections (List, Set, and Map).
5. Uses Dart operators and constants.

Lab-2:

Dart Programming Language
Learning Resources - Introductory Dart Video (Last 2 hrs.)

Practical Task –

Library Management System

Objective:
Develop a simple console-based library management system that manages a collection of books. The system should allow users to perform various operations while demonstrating the use of conditional statements, loops, functions, classes, inheritance, exception handling, and async programming.

Task Requirements:

1. **Book Class**:
    - Create a `Book` class with properties: `title`, `author`, `yearPublished`, and `isAvailable` (boolean).
    - Implement getters and setters for these properties.
    - Include a method to display book details.
2. **Library Class**:
    - Create a `Library` class that maintains a collection of `Book` objects.
    - Include a method to add a book to the library.

    - Include a method to borrow a book (mark it as not available).
    - Include a method to return a book (mark it as available).
    - Use a constructor to initialize the library with a predefined list of books.

3. **User Interaction**:

    - Use a loop to present a menu to the user with the following options:
        - Add a new book.
        - Borrow a book.
        - Return a book.
        - List all books.
        - Exit the system.
    - Use conditional statements to handle user inputs and navigate the menu.

4. **Switch-Case for Menu Operations**:

    - Use a switch-case statement to process menu selections.

5. **Handling Exceptions**:

    - Implement exception handling for cases like attempting to borrow a book that is not available or returning a book that was not borrowed.

6. **Inheritance**:

    - Create a subclass `EBook` that inherits from `Book` and adds a property `fileSize`. Override the method that displays book details to include the file size.

7. **Static Methods**:

    - Implement a static method in the `Library` class that keeps track of the total number of books in the library.

8. **Abstract Class**:

    - Create an abstract class `User` with an abstract method `displayUserType()`. Create a subclass `Member` that implements this method.

9. **Use of Async and Await**:

    - Simulate an asynchronous process for listing books with a method in the `Library` class that returns a `Future<List<Book>>` and uses `await` to simulate a delay.

Week 2 & 3: Introduction to Flutter, Widgets and Layouts

Lab-1 :

Learning Resources - https://www.youtube.com/watch?v=9uq2HpIq2XY

Practical Task - Build a simple Hello World app in Flutter.

Topics:
Widget Hierarchy - Root Widget, Child Widgets, Parent-Child Relationships
Composition - Combining Widgets, Nested Layouts
Widget Types - Stateless Widgets, Immutable Properties, Rendering Pipeline

Learning Resources -
https://www.youtube.com/watch?app=desktop&v=gOZrczjHF6g
https://docs.flutter.dev/tools/devtools/inspector
https://medium.com/flutter-community/flutter-widgets-row-column-flex-the-whole-picture-a648cd6e6904
https://www.youtube.com/watch?v=g2E7yl3MwMk
https://www.youtube.com/watch?app=desktop&v=gOZrczjHF6g
https://medium.com/@bosctechlabs/building-dynamic-layouts-with-flutter-flex-da5576180882

Practical Task -
1) Create an App with list, grids and Scrolling for To-do list App.

Lab-2:

Topics:
Stateful Widgets - Mutable State, Life Cycle Callbacks
Common Widgets (Text, Image, Button)
Layout Widgets (Row, Column, Stack, Expanded, Flex)

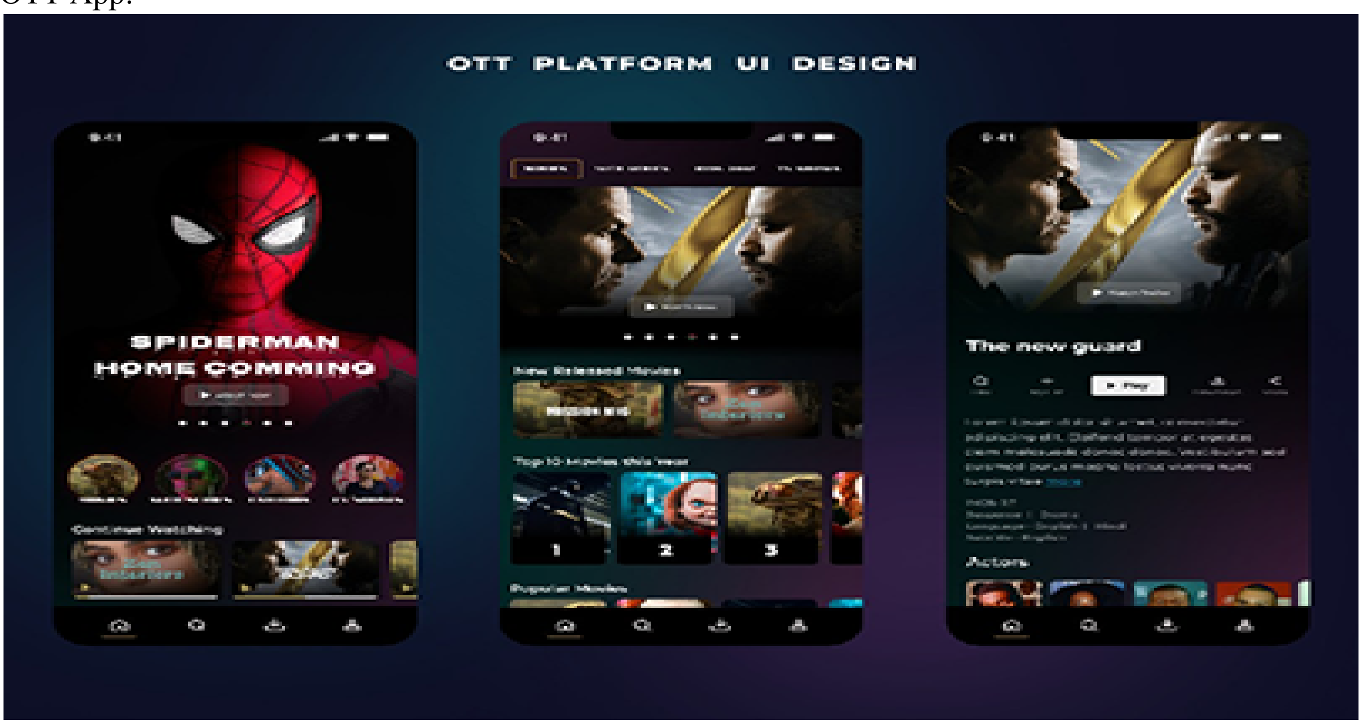
Learning Resources -
https://www.youtube.com/watch?app=desktop&v=gOZrczjHF6g
https://docs.flutter.dev/tools/devtools/inspector
https://medium.com/flutter-community/flutter-widgets-row-column-flex-the-whole-picture-a648cd6e6904
https://www.youtube.com/watch?v=g2E7yl3MwMk
https://www.youtube.com/watch?app=desktop&v=gOZrczjHF6g
https://medium.com/@bosctechlabs/building-dynamic-layouts-with-flutter-flex-da5576180882

Practical Task -
1) Create an App with Multiple Widgets and Layouts for News App and OTT app

Output:

News App: https://docs.flutter.dev/resources/news-toolkit

OTT App:



Week-4

Lab 1 & 2: Implementing a Login Screen in Flutter Application

Learning Resources

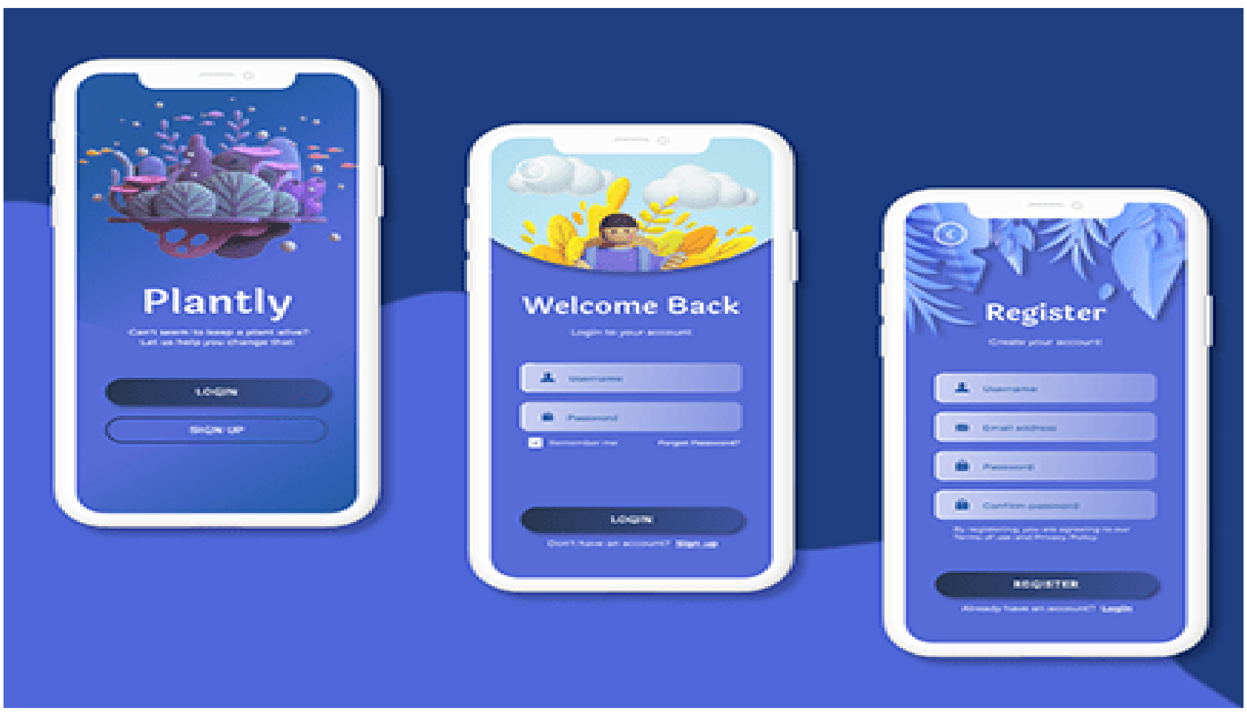
- [Flutter Login Tutorial]()
- [Firebase Authentication Guide]()

Practical Task:

- Create a login screen using `TextField` widgets for username and password input.
- Integrate Firebase Authentication for user login and registration.

Topics Covered:

- Widgets: Text, Button, Layout
- Handling User Input and Forms
- Firebase Authentication Integration



Week-5

Lab 3 & 4: Build a Shopping Cart App with Flutter

Learning Resources

- [State Management in Flutter]()

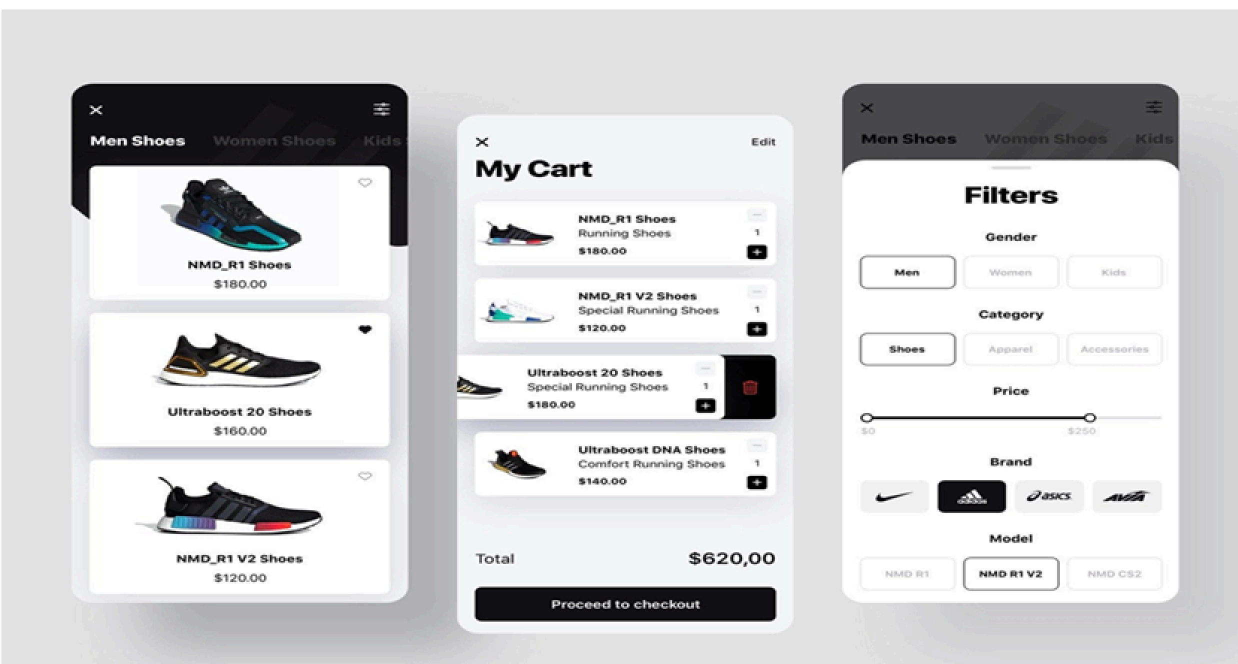
Practical Task:

- Build a shopping cart app with product listings using `ListView` and `GridView`.
- Use `setState` to manage cart items dynamically.

Topics Covered:

- Stateful Widgets and State Management
- Navigation and Routing
- User Interaction with Lists

output:



Week-6

Lab 5 & 6: Create an E-Book App Using Flutter

Learning Resources

- Validating Input
- Custom Widget Development
- Navigation and Routing

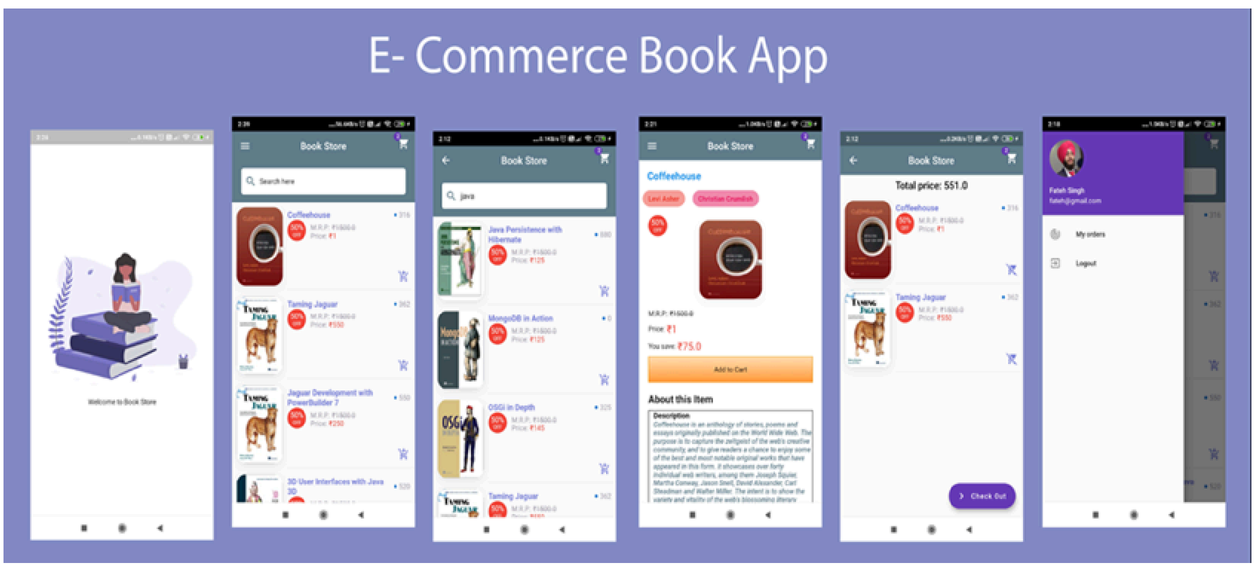
Practical Task:

- Create an E-Book app with sections for book categories and a simple book reader UI.
- Implement navigation between categories and book details.

Topics Covered:

- Custom Widgets for Book Items
- Navigation and Routing
- Validating Input

output:



Week-7

Lab 7 & 8: Create a Video Player App Using Flutter

Learning Resources

- https://docs.flutter.dev/cookbook/plugins/play-video
- Media Handling in Flutter

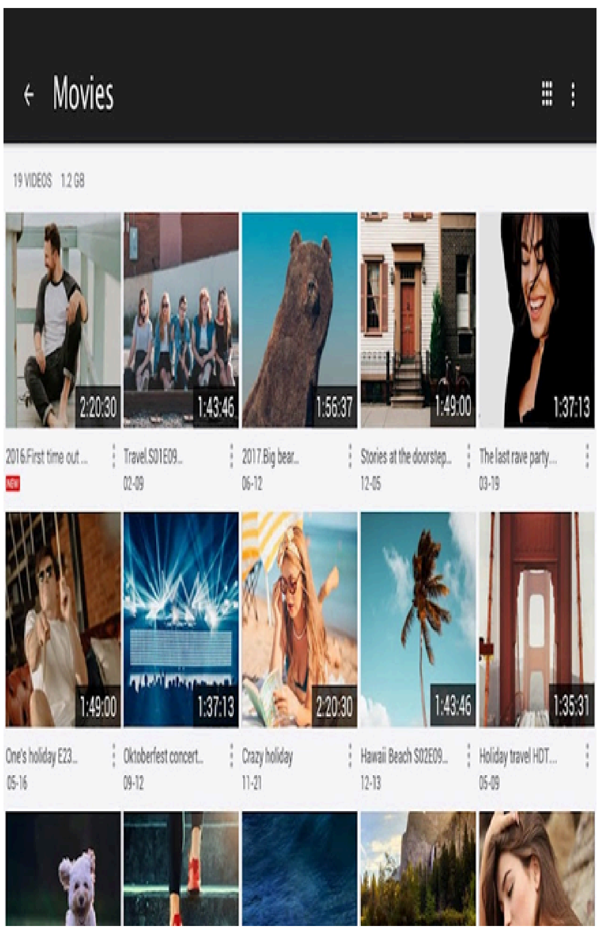
Practical Task:

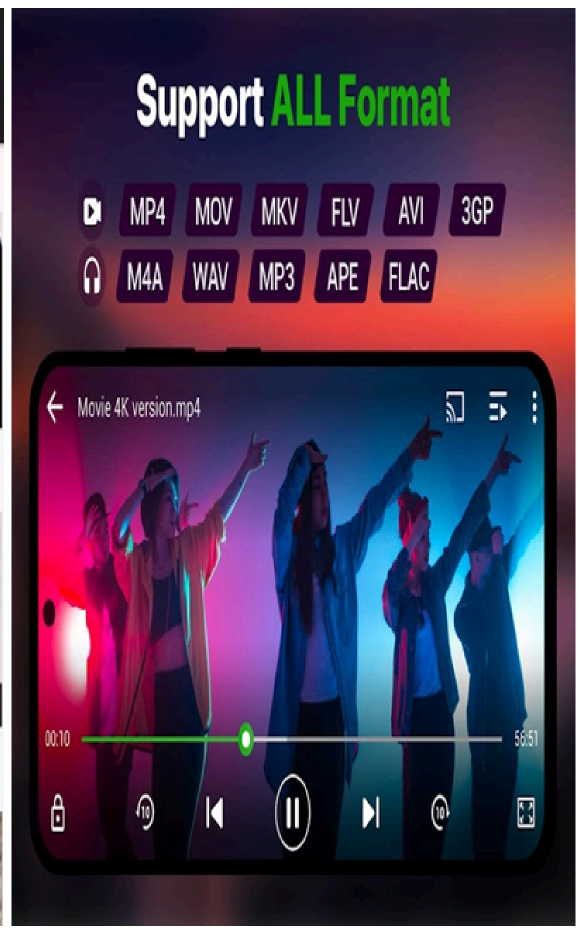
- Build a video player app.
- Add controls for play, pause, and seek functionality.

Topics Covered:

- Integrating External Plugins
- Media Query and Responsive Design
- Gesture Detection for Video Controls

Output:





Week-8

Lab 9 & 10: Create a Photo Editing App Using Flutter

Learning Resources

- https://www.youtube.com/watch?v=Hxh6nNHSUjo
- Handling Images in Flutter

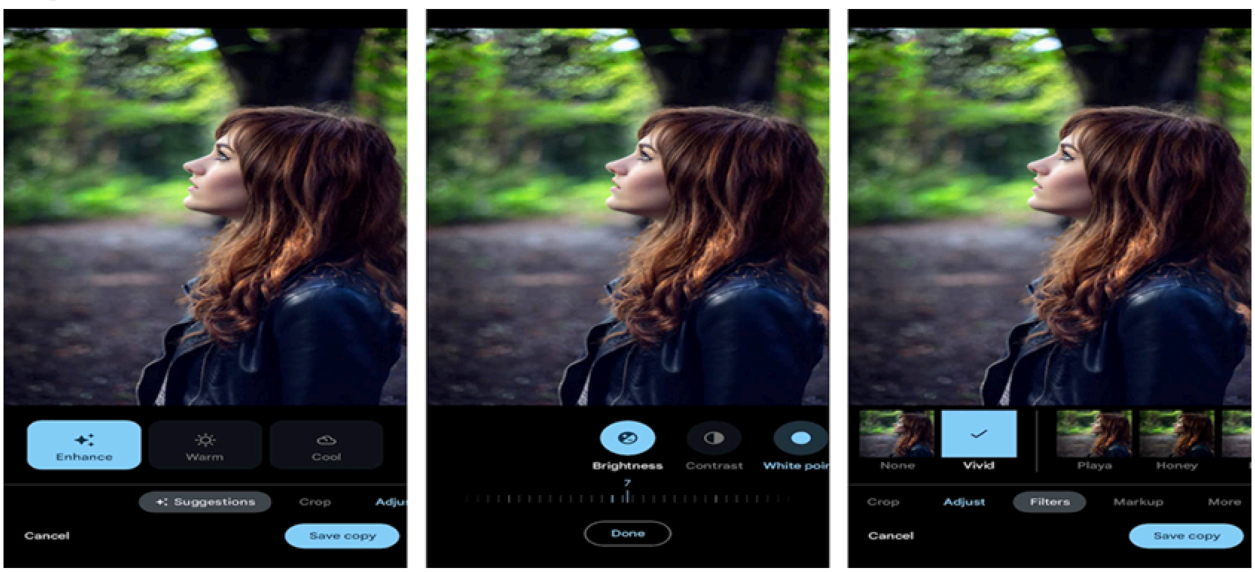
Practical Task:

- Create a photo editing app with features to crop, rotate, and apply filters to images.
- Use editing functionalities.

Topics Covered:

- Plugin Integration for Photo Editing
- Handling Images with Widgets
- Stateful Widgets for Dynamic Editing

Output:



Week-9

Lab 11 & 12: Create a Music Player App Using Flutter

Learning Resources

- https://pub.dev/documentation/just_audio/latest/
- Audio Handling in Flutter

Practical Task:

- Develop a music player.
- Add controls for play, pause,next, previous and stop with track progress.

Topics Covered:

- State Management for Track Controls
- Styling Widgets for Playlist UI

Output: